

Weinberg-operator coefficient

$$T3 - d3 - d5 - F4, \alpha = 0 : \quad S_1 = (3, 0), \quad S_2 = (5, 1), \quad F = (4, \frac{1}{2})$$

$$C_5 = \frac{2}{3} \left(-\frac{\sqrt{15} h M_F^3 M_{S2}^2 \lambda_{T3} \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{y_{1,i1,r1} y_{2,i2,r1}}}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} + \frac{\sqrt{15} h M_F^3 M_{S1}^2 \lambda_{T3} \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{y_{1,i1,r1} y_{2,i2,r1}}}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} - \frac{\sqrt{15} h M_F M_{S1}^2 M_{S2}^2 \lambda_{T3} \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{y_{1,i1,r1} y_{2,i2,r1}}}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} + \frac{\sqrt{15} h M_F M_{S1}^2 M_{S2}^2 \lambda_{T3} \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{y_{1,i1,r1} y_{2,i2,r1}}}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \right)$$